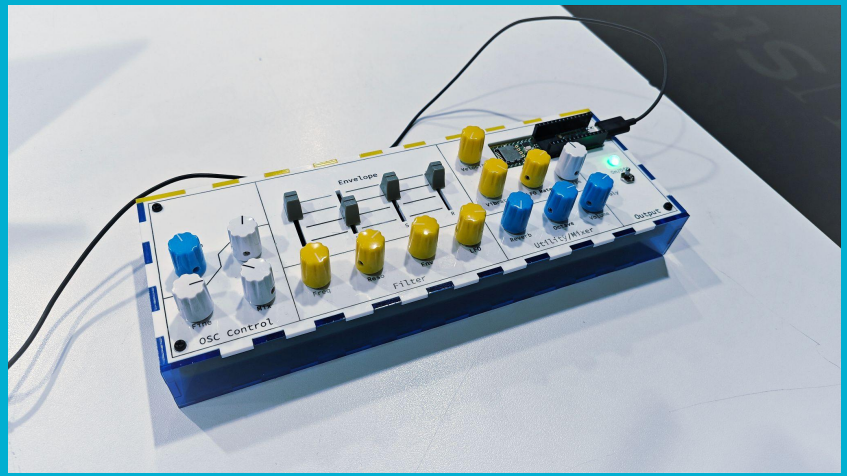
TeensyLab V2



Building a Digital Guitar-to-Synth System Using Max, JUCE and Teensy

Michael Xie, Spring 2024 Capstone Presentation

Advisor: Prof. Steven Litt & Prof. Paul Geluso

Introduction
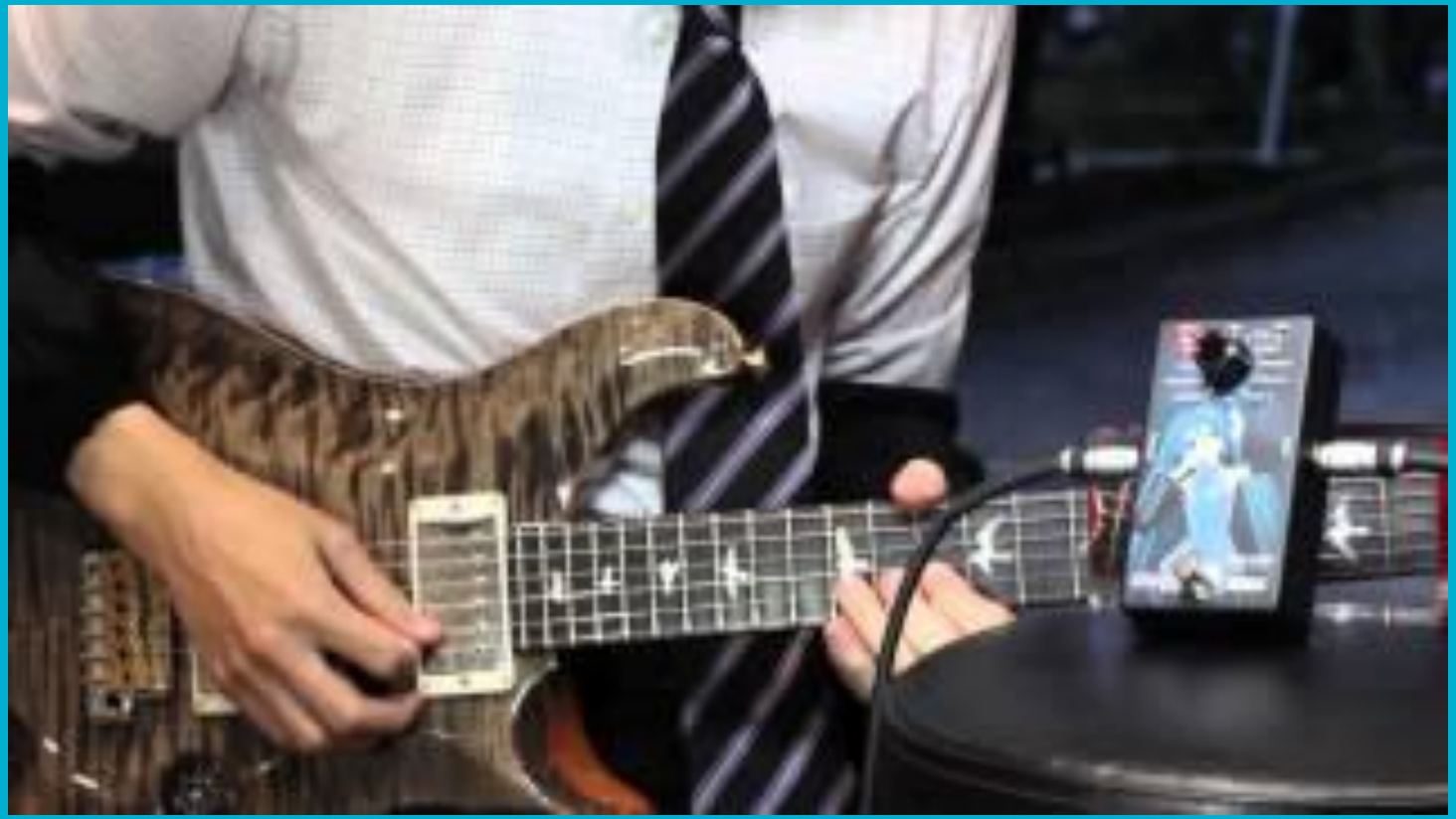
- Goal: Using the tools learned, build a system mimic the guitar-to-synth pedals. The first inspiration is MIKU KORG, a guitar-to-vocal pedal.
- Miku Korg, Mooer E7, BOSS SY-1







Introduction



Introduction

- Method A: F0 detection/estimation -> midi event. YIN algorithm based on autocorrelation method.
- Method B: Transformation in the frequency domain.



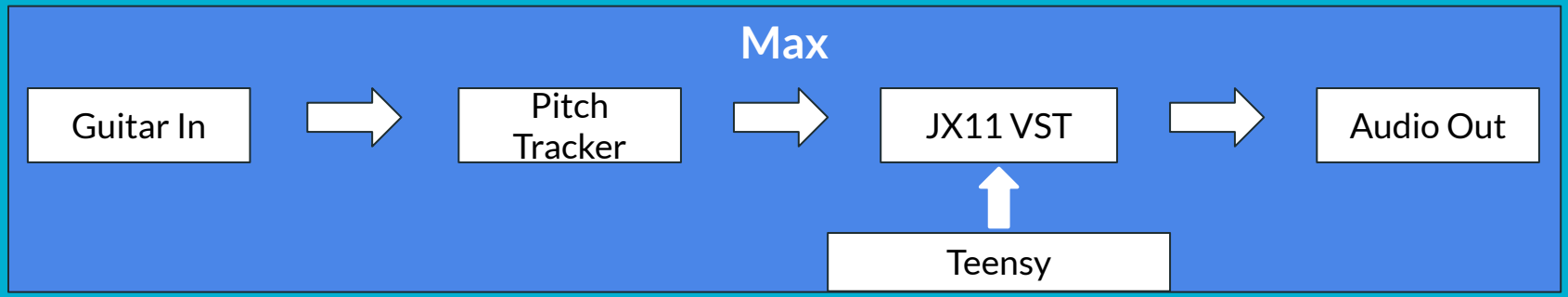




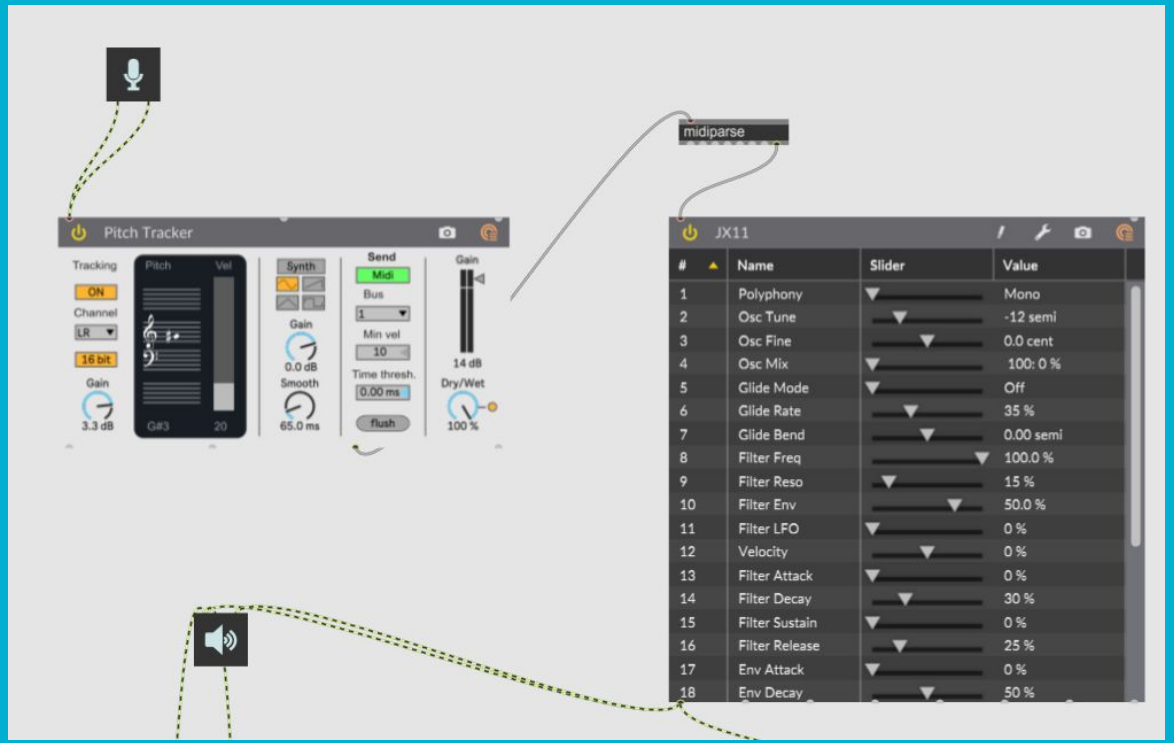
Project Description

3 Parts:

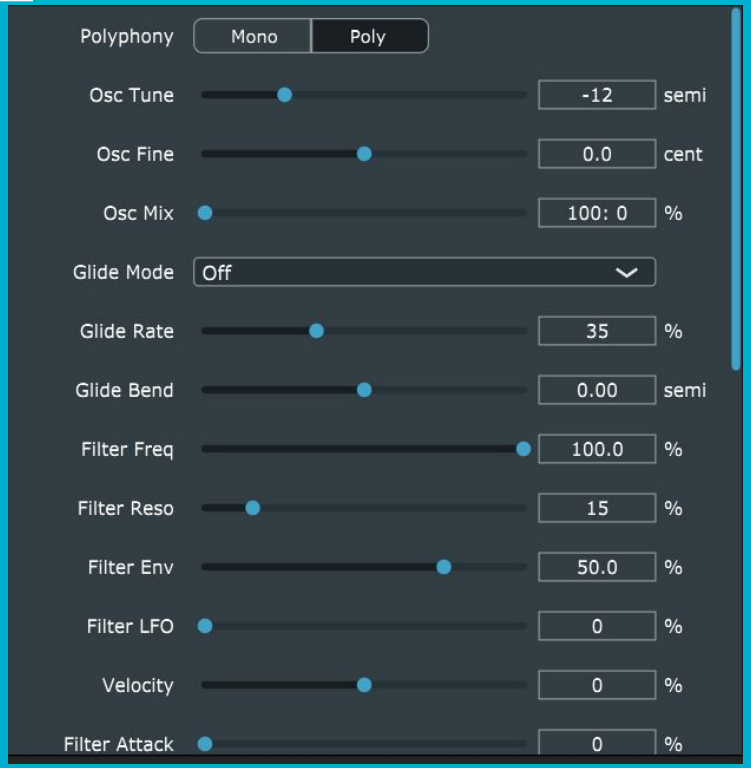
- Max Environment: Pitch Tracker Module, host for the vst.
- The VST plugin: JX11 plugin, a 2 voice sawtooth virtual analog synth.
- Hardware teensy control interface: Midi controller built with Teensy, a prototype with Printed Circuit Board and laser-cut box.

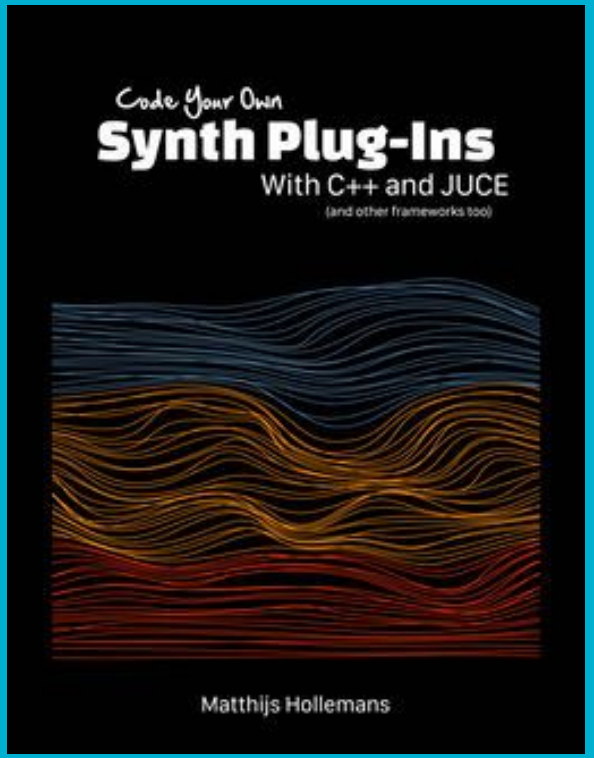


Max Environment



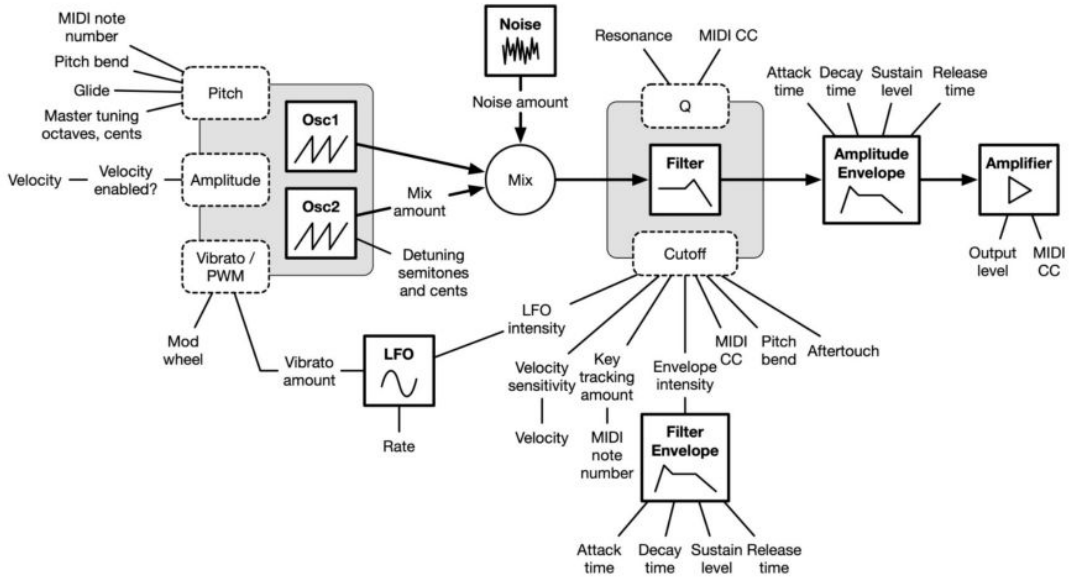
The JX11 VST Plugin





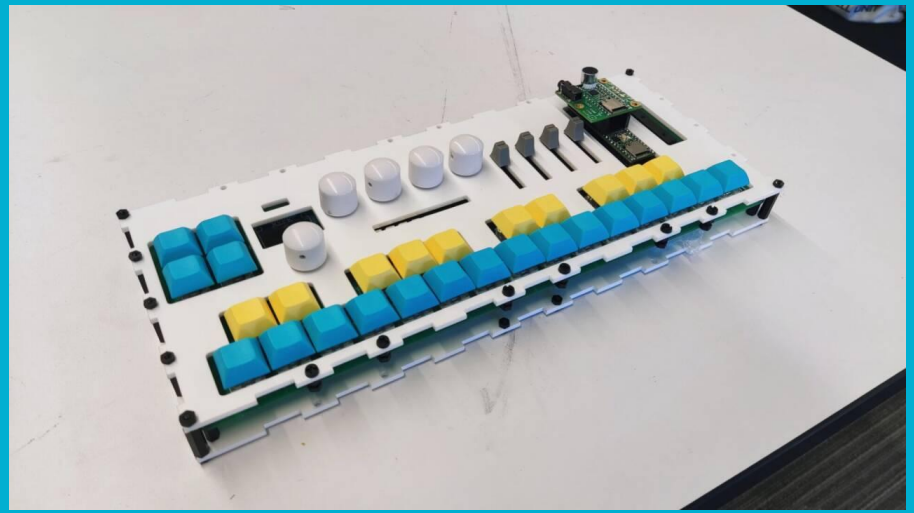
The JX11 VST Plugin

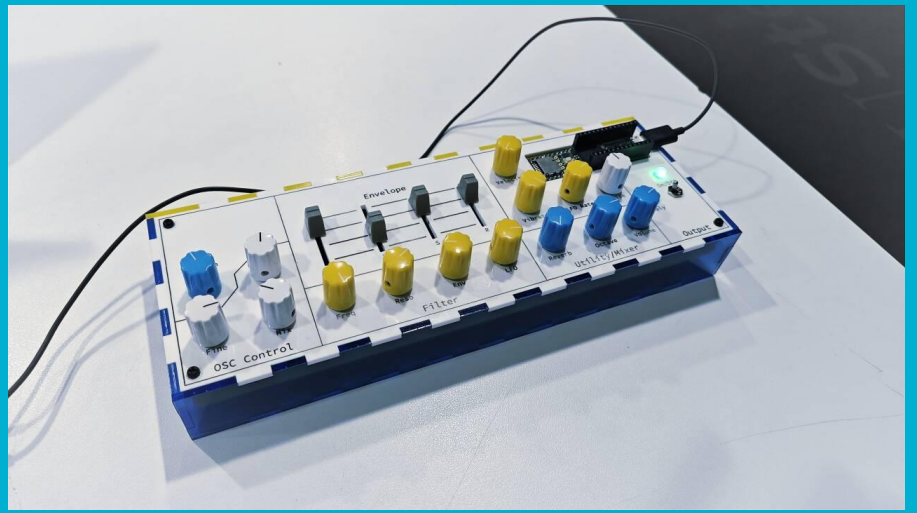
Here is the completed schematic of JX11 with all the pieces that make up the synth:



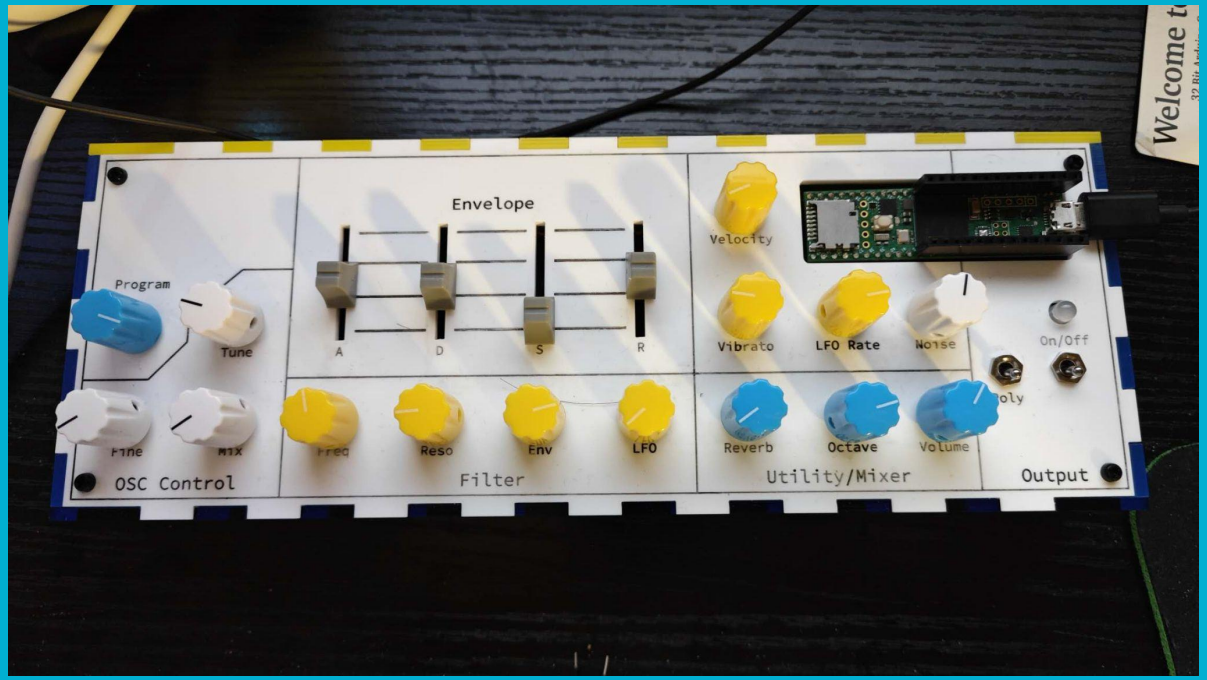
The full schematic of the synth

Control Interface with Teensy



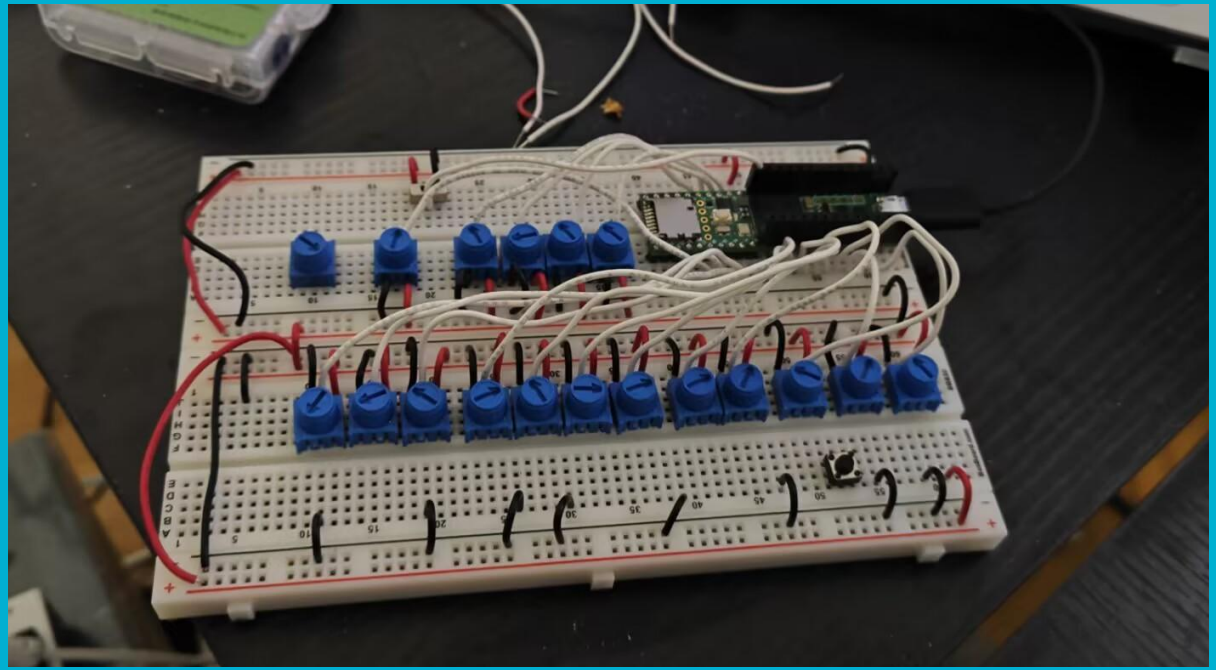


Control Interface with Teensy



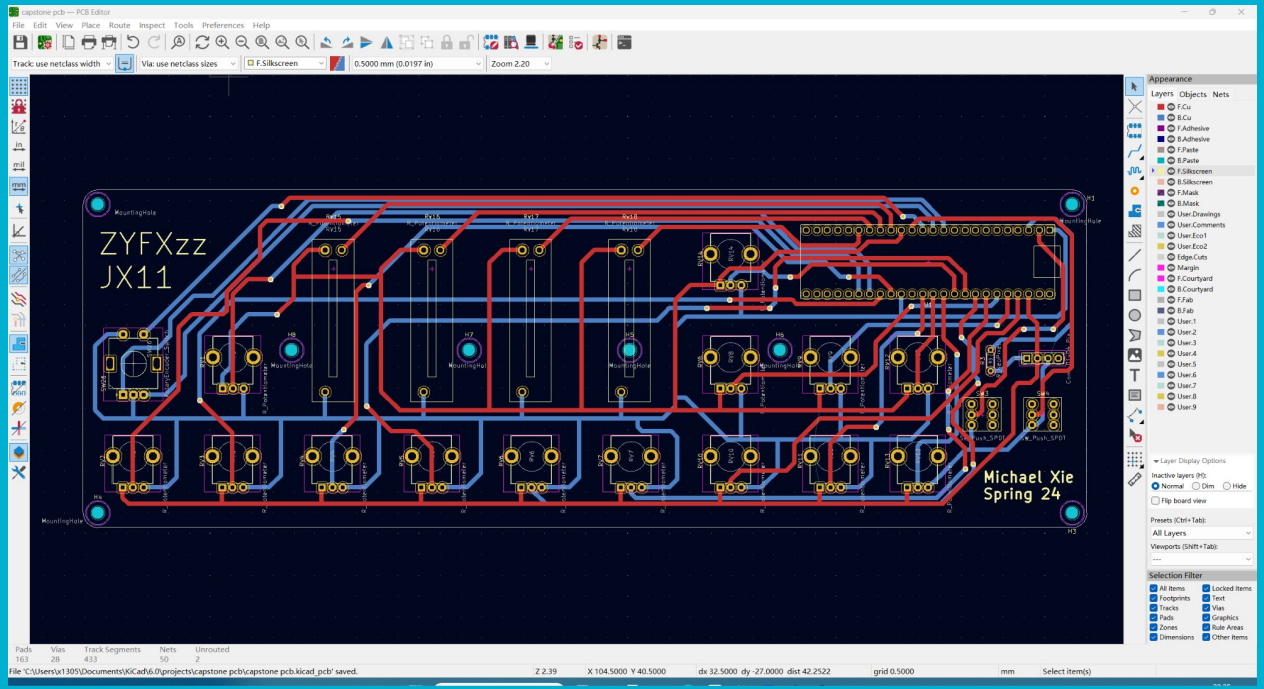
- Encoder * 1
- Potentiometer * 18
- 2 Switches
- Neopixel LED

Control Interface with Teensy



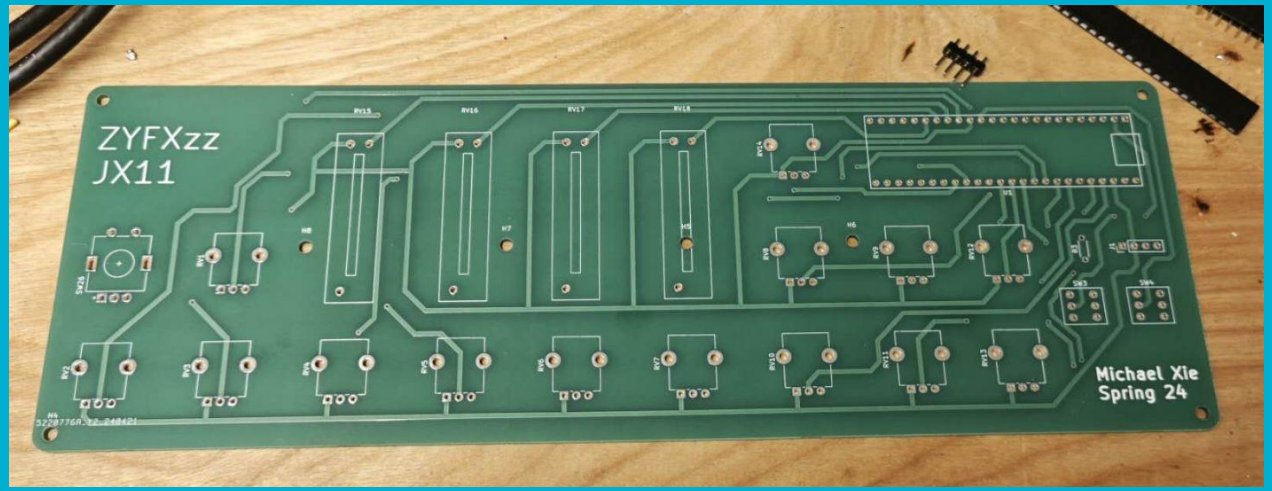
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Control Interface with Teensy



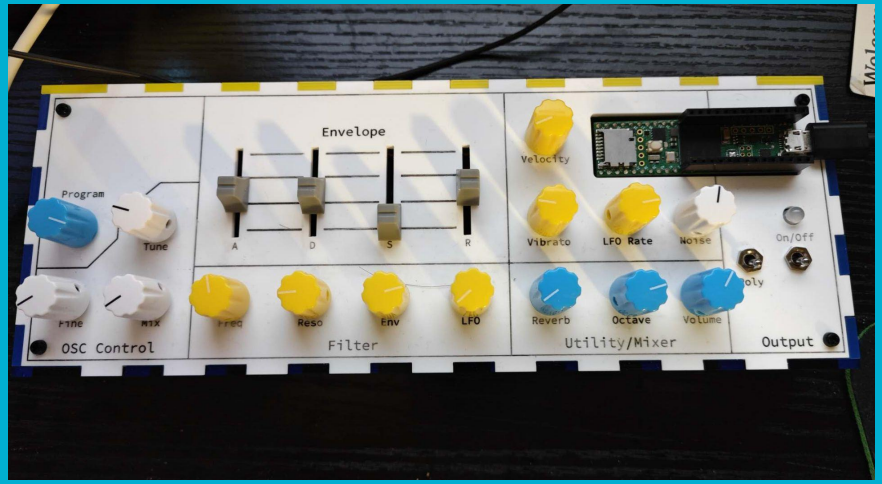
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Control Interface with Teensy



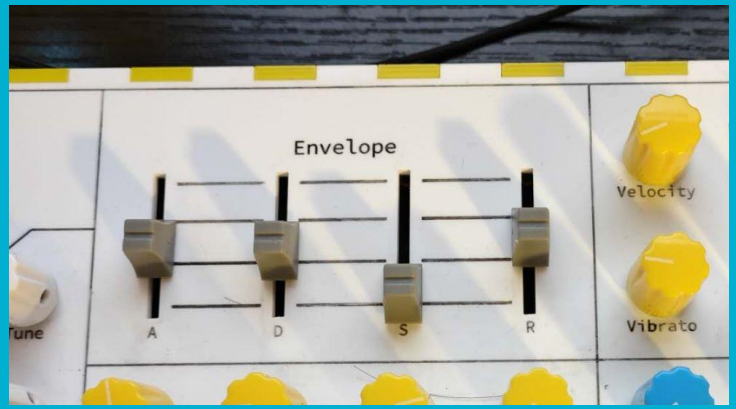
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Control Interface with Teensy





Control Interface with Teensy





Demo



Future Work

- Add sample-based synthesis, which was expected to accomplish when proposing the project.
- Improve pitch tracking: using other tools available, midi debounce mechanism, distinguish between a new note event and vibrato/pitch bending.
- Expand max patch, put in midi effects. A note in, a chord out.
- Machine Learning Timbre transfer, DDSP from Google Magenta. With ML, DDSP can transform bird singing to violin playing.

Conclusion

- A better understanding of structure/life-cycle of an audio plugin
- A continue practice of electronics
- A good summary of my study in the undergrad program
- A stepping stone for future work in the master

Thank you!

Reference

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